

XIAOGE ZHANG, PhD

Cambridge, UK | **Full Right to Work in the UK (No sponsorship required)**

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Professional Profile

Quantitative scientist and modeller (PhD, York) with 7+ years' experience in modelling, causal inference research, statistical analysis, and technical reporting, now applying this background to HEOR and HTA. Developed hands-on experience in decision-analytic modelling, evidence synthesis, and NICE-style appraisal methods through an independent technical portfolio reconstructing published NICE appraisals. Combines academic rigour with practical model-building and validation experience in R, Stan, and Excel.

Technical Competencies

- **Economic Evaluation:** Hands-on experience building and validating NICE-style Partitioned Survival Models (PSM) and multi-state Markov models through independent portfolio work; familiar with structural assumptions including treatment waning, cure fractions, ONS-linked background mortality, PSA/DSA, and threshold analysis.
- **Evidence Synthesis:** Portfolio-based experience reconstructing evidence networks and implementing Bayesian Network Meta-Analysis (NMA), including population-adjusted indirect comparisons and ML-NMR concepts; familiar with parametric survival extrapolation methods including Weibull, Gompertz, and spline-based approaches.
- **Real-World Evidence:** Academic research experience in causal inference, observational study design, and applied statistical modelling; familiar with UK real-world data sources relevant to NICE submissions, including SACT and CPRD, and with propensity score methods for addressing baseline imbalance.
- **Technical Toolkit:**
 - **R:** flexsurv, survival, meta, netmeta, tidyverse; experienced in reproducible scripting and output generation.
 - **Stan:** Hierarchical Bayesian models, MCMC diagnostics (R-hat, ESS).
 - **Excel:** Decision-analytic model building, Solver-based calibration, automated PSA/DSA dashboards.
 - **MS Office:** technical reports, slide decks, and stakeholder-facing outputs.

Independent HEOR Technical Portfolio

A self-directed technical repository applying quantitative methods to NICE/SMC evidence standards.

Weblink: xgzhang.com (Models, Code, and Visual Analytics)

Case Study 1: Oncology Model Validation & Remediation (Excel/R)

2026

- **Focus:** Technical validation and remediation of an NICE-style **3-State PSM** for oncology, ensuring structural consistency and biological plausibility for NICE submission.
- **Remediation:** Resolved time-unit mismatches causing survival crossovers; implemented a hazard-chaining algorithm in R to eliminate discontinuities at the trial-to-extrapolation boundary.
- **Decision Relevance:** Linked model corrections to cost-effectiveness implications, including ONS-linked background mortality, subsequent therapy costs, and PAS discount thresholds.
- **Deliverable:** reproducible remediation workflow and automated PSA/DSA dashboard suitable for technical review.

Case Study 2: Evidence Synthesis & External Control Arms (R/Stan)

2025–2026

- **Focus:** Reconstructing evidence networks and External Control Arms (ECA) to quantify treatment effects in single-arm oncology studies (e.g., TA1013, TA1017) lacking direct comparators.
- **Analytics:** Applied population-adjusted indirect comparison methods and propensity-based weighting to address baseline imbalance; developed Bayesian hierarchical models in Stan with full MCMC diagnostics (R-hat, ESS).
- **Technical Guide:** Built an eight-part NMA technical guide mapping key evidence synthesis methods, including Bayesian NMA, NMR, survival NMA, CNMA, IPD-NMA, MAIC, STC, and ML-NMR, with notes on assumptions, implementation logic, and HTA relevance. [Weblink](#)
- **Deliverable:** Generated reproducible evidence synthesis outputs, sensitivity analyses, and technical summaries suitable for methodological review.

HTA Appraisal Review & Methodology Repository

2025–Present

- **Scope:** Systematic review of 20+ NICE appraisals, examining methodological choices and their impact on cost-effectiveness conclusions across oncology and rare disease submissions.
- **Repository:** Developed 50+ technical reference cards covering effective sample size (ESS) shrinkage, treatment waning, and non-linear utility mapping—providing a structured knowledge base for model development and quality assurance.

Professional Experience

Lecturer → Associate Professor (Quantitative Economics & Modelling)

2018 – 2025

Southwestern University of Finance and Economics

Sichuan, China

- **Scientific Writing & Research Delivery:** Principal writer for a sub-project within a National Social Science Fund Major Grant (2020–2023); produced structured technical reports synthesising quantitative evidence under competitive peer review and institutional quality standards.
- **Quantitative Project Supervision:** Supervised 15 Master's students through full research cycles—from study design and analytical methodology to final reporting—managing concurrent projects with structured feedback and quality review at each stage.
- **Statistical Communication:** Led bilingual curriculum design for quantitative methods courses; consistently recognised for translating complex modelling concepts into accessible, structured outputs for diverse audiences.

Graduate Teaching Fellow (Statistics & Economics), University of York

2013 – 2017

- Delivered postgraduate and undergraduate teaching in Mathematics, Statistics, and Economics during PhD; awarded Teaching Excellence Awards in two consecutive years for clarity in communicating quantitative methods.

Education

University of York, Ph.D. in Economics

York, UK | 2013 - 2018

University of Sheffield, M.Sc.

Sheffield, UK | 2011 - 2012

Selected Publications

Economics Letters (2023): Indirect Inference framework for bias calibration in irregular-interval panel data models

Pacific-Basin Finance Journal (2022): Firm risk-taking under monetary policy uncertainty

Additional Information

- **Languages:** English (Full Professional Proficiency); Mandarin (Native).
- **Visa Status:** Full Right to Work in the UK (No sponsorship required).
- **Certification:** Associate Fellow (AFHEA) of Advance HE, UK (awarded 2025).